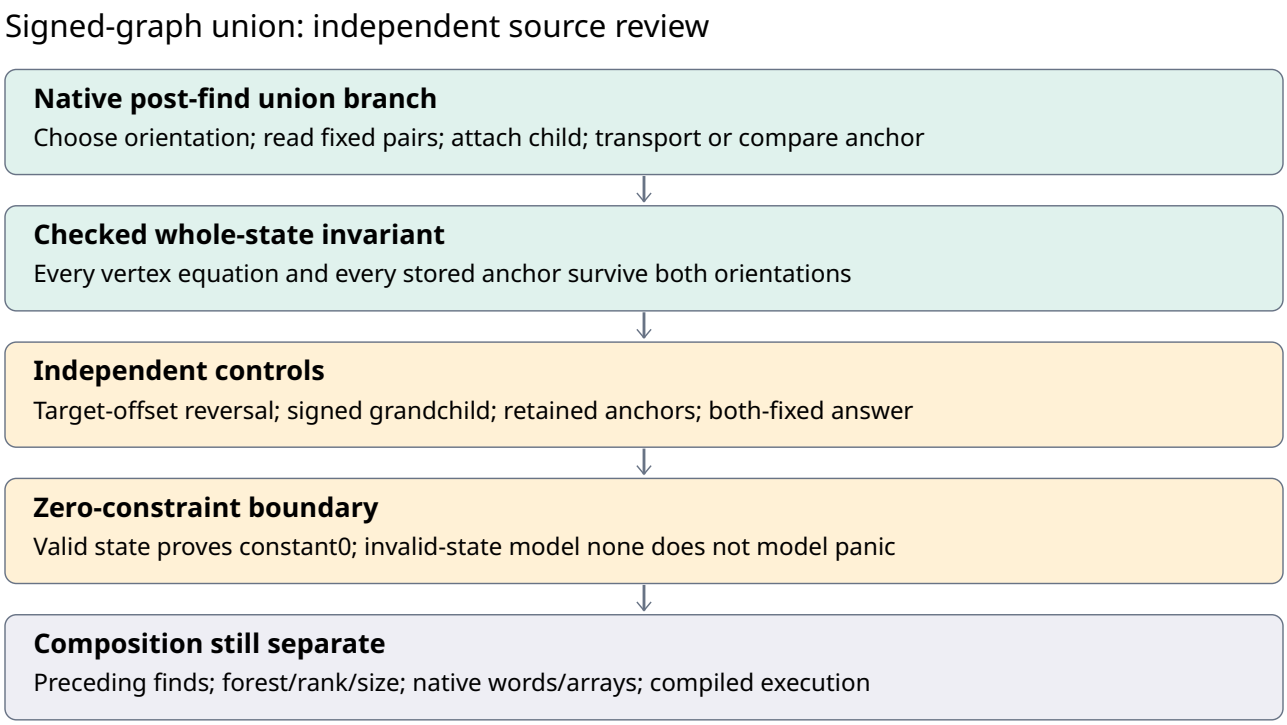
Independent review of signed-graph union mutation



pencil-graph-union-review-20261101 | 5 main + 8 producer + 12 reviewer | no science

Five main declarations, eight producer controls and twelve independent reviewer controls passed fresh replay, with twenty-five exact named axiom readbacks. The reviewed model preserves the full parent/potential representation and every stored anchor under either union orientation. It also proves that a both-fixed nonzero constraint returns the modeled target and that a zero coefficient is consistent under the invariant. The result is validated at its explicitly post-find, decoded-field state scope.

Inputs and review sequence. The mathematical source files, helper and PLAN were extracted from published commit3d48e8e3ee in PR3056, union-division-runtime/union, and matched the producer's private files byte-for-byte. All620 producer manifest entries matched. The prospective coordinator union joint is copied in coordinator-review-plan.json. Native three_leads.rs1658-1712 was inspected before Union.lean. The local PLAN.md is an execution note written after initial source/proof reading, not a retroactively prospective plan. This reviewer authored the older GraphAlgebra component and previously reviewed find/cycle source; no fully blind history is claimed.

Literal source correspondence. The native distinct-root branch derives s=-cu*cv, aa=cu*a and bb=cu*b. The size-selected reverse branch keeps s and changes offsets to -s*aa and -s*bb, matching reverse(unionPotential). Native code reads both fixed pairs before changing child parent/potential, then increments parent size. If the child is fixed it transports each coordinate by pot.s*(old-pot.offset), assigns only an unfixed parent, or compares with an existing parent anchor. joined and result express these parent/potential/fixed mutations and the result from the old fixed pairs. The original child anchor is retained, exactly as in the native source.

Global invariant. Valid requires every vertex equation v[i]=eval(pot[i],v[parent[i]],k) and every stored anchor equation, including anchors on nonroots. join_preserves updates the child's local equation and proves every other vertex equation still holds, since the semantic values v are unchanged. Descendants retain their own parent/potential equations. Transferred parent anchors follow from the signed transport theorem; all retained anchors remain valid. forward_global and backward_global derive the necessary endpoint equation from the reduced edge and unit-square signs before applying the global invariant theorem. They do not assume the global conclusion or merely certify endpoint equality.

Both-fixed and zero-constraint boundary. join_result uses the existing and transported parent anchors to identify any returned nonzero-coefficient candidate with k. both_fixed_zero_consistent proves that a zero difference coefficient forces the difference constant to be zero under Valid and the true edge. Native constraint asserts that constant is zero. The model's constraint function returns none whenever its coefficient is zero, even for invalid states that would trigger the native assertion. A reviewer counterexample exposes this boundary: transport gives(9,-1), an inconsistent existing

anchor(8,-1) gives coefficient0 and constant-1, model result none, and the prestate Valid premise is disproved. Thus the result function alone is not a model of native panic behavior on arbitrary invalid inputs; the checked valid-state theorem excludes the case.

Independent controls. A four-node rational fixture has roots0,1, child2->0 and grandchild3->2, signed and target-dependent offsets, plus stored anchors on0 and3. At k=2 its values are3,7,4,6. The forward edge uses cu=1, cv=-1, a=-8, b=2, giving potential(+1,-8,2); reversal changes offsets to(+8,-2). Both orientations instantiate the global theorems. The grandchild equation and parent link remain unchanged; the child anchor(1,1), transported parent anchor(9,-1) and descendant anchor(2,2) are all checked. Both-fixed anchors produce answer2; a consistent zero constraint returns none. Incorrect reversal of the target offset violates representation. These complement the producer's negative-sign orientation and descendant controls.

Machine evidence. A fresh invocation compiled Find, Cycles, Union, Controls, Readback, Reviewer and Reviewer-Readback, all at native0/gate0 with unchanged bindings. accept.py requires this exact module sequence, exact source bytes and all25 named readbacks with only propext, Classical.choice and Quot.sound. A deliberate False theorem was rejected at native1/gate1 after its dependencies compiled. An earlier negative-control invocation omitted Union's compilation and failed on the missing module; it is retained and labeled as dependency failure, not the False control. Producer failures remain in the original packet. The snapshot and cached dependency binaries retain their stated trust boundary; no independent kernel binary rechecker was run.

Post-find and native boundaries. State begins after both finds and assumes a true reduced edge, a valid prestate and unit-square decoded signs. Rootness, distinct roots and size comparison select the native branch, but the equation-preservation theorem itself does not need them. Consequently it does not establish a forest/rank invariant or the correctness of union-by-size metadata; those are needed to compose repeated unions with the earlier find termination theorem. Integration of the two preceding mutating finds, node allocation, Vec bounds, i8/u64 interpretation, counter/size overflow, panic execution and compiled Rust remains outside this packet. No Jac(C)-to-scalar map is constructed here.

Review outcome. Machine status passed; source fidelity faithful_scoped; review outcome validated_scoped for global equation/anchor preservation and valid-state constraint correctness. Both the stronger algebraic generality and the concrete native boundaries are explicit. PDF and diagram accompany the structured review. No scientific execution, cloud work, repository edits or producer-snapshot changes occurred.